KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI
POLITEKNIK NEGERI MALANG
JURUSAN TEKNOLOGI INFORMASI
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Mata Kuliah	:	**PEMROGRAMAN BERBASIS FRAMEWORK (PBF)**
Program Studi	:	D4 – Teknik Informatika
Semester	:	6

Kelas	:	TI-3B
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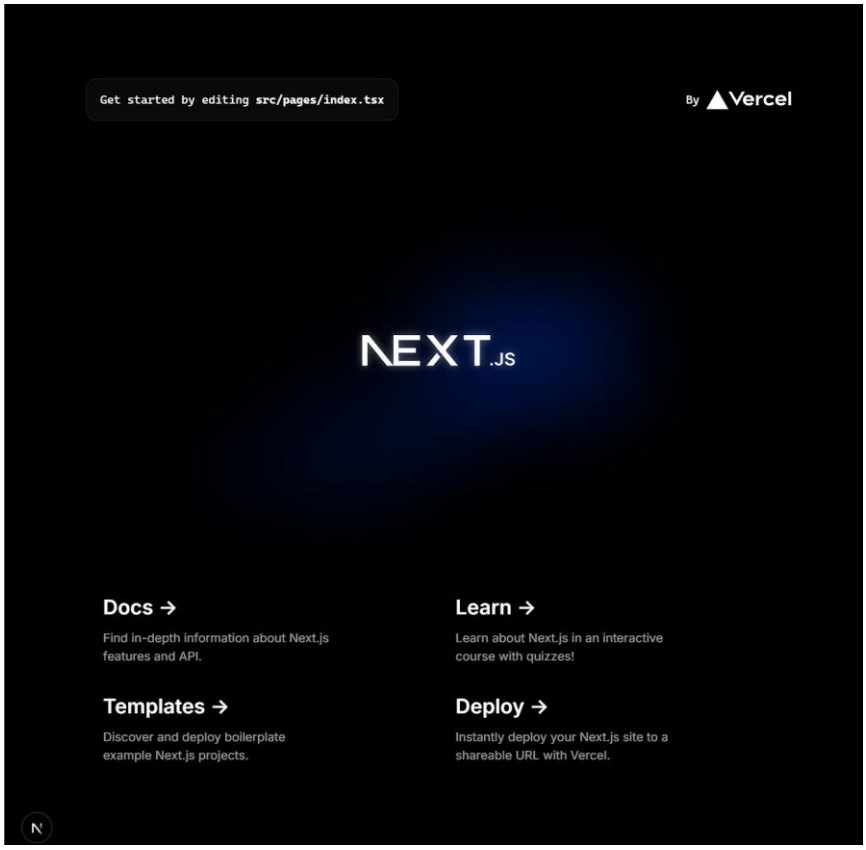
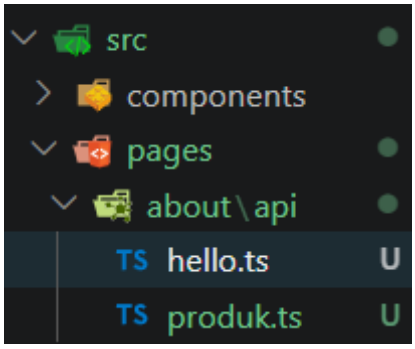
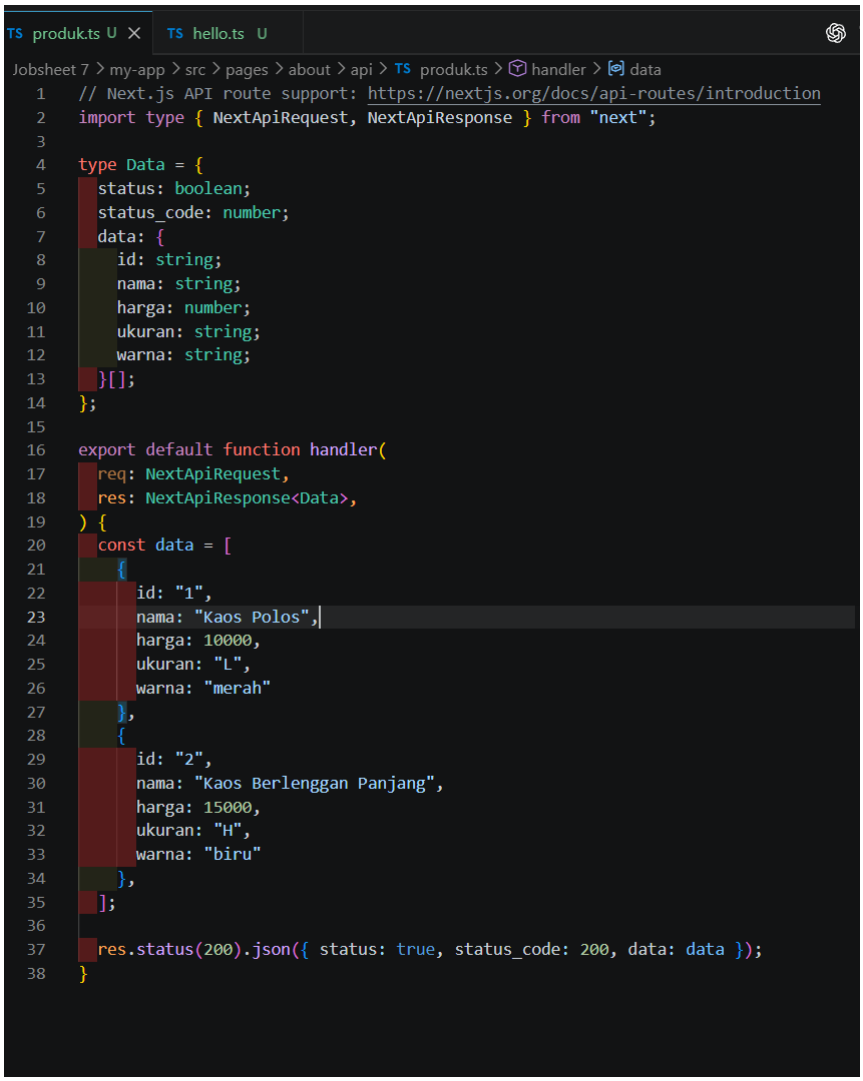
Nama	:	Raudhil Firdaus Naufal
Jobsheet Ke-	:	7

Laporan Jobsheet

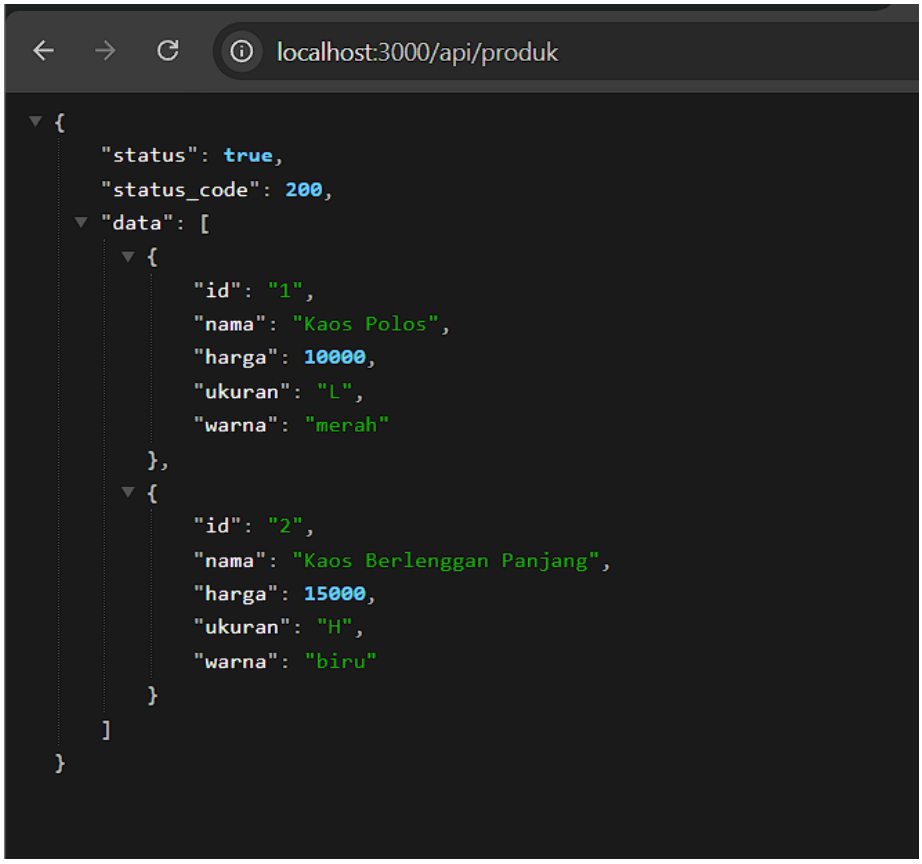
Langkah 1 - Menjalankan Project

Langkah	Jawaban/Deskripsi
1	npm run dev 

Langkah 2 - Membuat API Produk

Langkah	Jawaban/Deskripsi
1	Buat file pad pages/api/produk.js 
2	Tambahkan data statis: 

```ts
// Next.js API route support: https://nextjs.org/docs/api-routes/introduction
import type { NextApiRequest, NextApiResponse } from "next";

type Data = {
  status: boolean;
  status_code: number;
  data: {
    id: string;
    nama: string;
    harga: number;
    ukuran: string;
    warna: string;
  }[];
};

export default function handler(
  req: NextApiRequest,
  res: NextApiResponse<Data>,
) {
  const data = [
    {
      id: "1",
      nama: "Kaos Polos",
      harga: 10000,
      ukuran: "L",
      warna: "merah"
    },
    {
      id: "2",
      nama: "Kaos Berlenggan Panjang",
      harga: 15000,
      ukuran: "H",
      warna: "biru"
    },
  ];

  res.status(200).json({ status: true, status_code: 200, data: data });
}
```

3	Akses: http://localhost:3000/api/produk  Berhasil menampilkan API

Langkah 3 – Fetch Data API di Frontend

Langkah	Jawaban/Deskripsi
1	Buka pages/product/index.tsx Lalu Modifikasi

TS produk.ts U index.tsx U ×

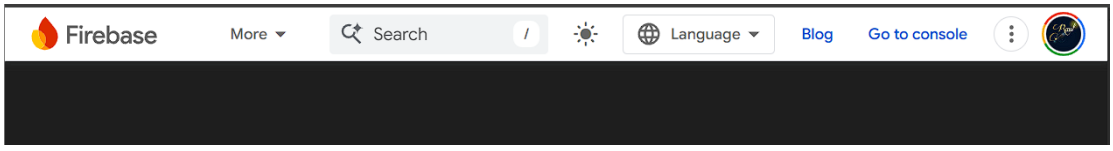
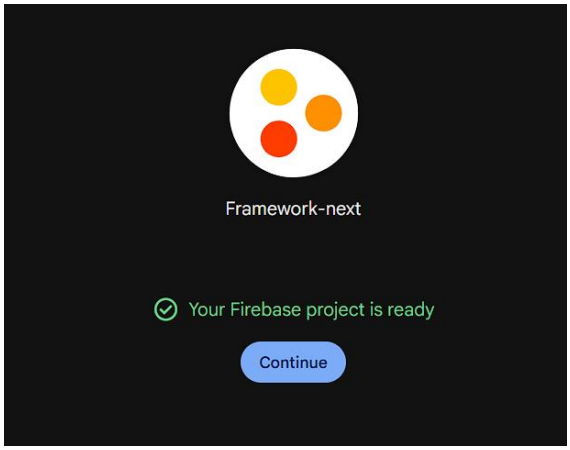
Jobsheet 7 > my-app > src > pages > produk > index.tsx > default

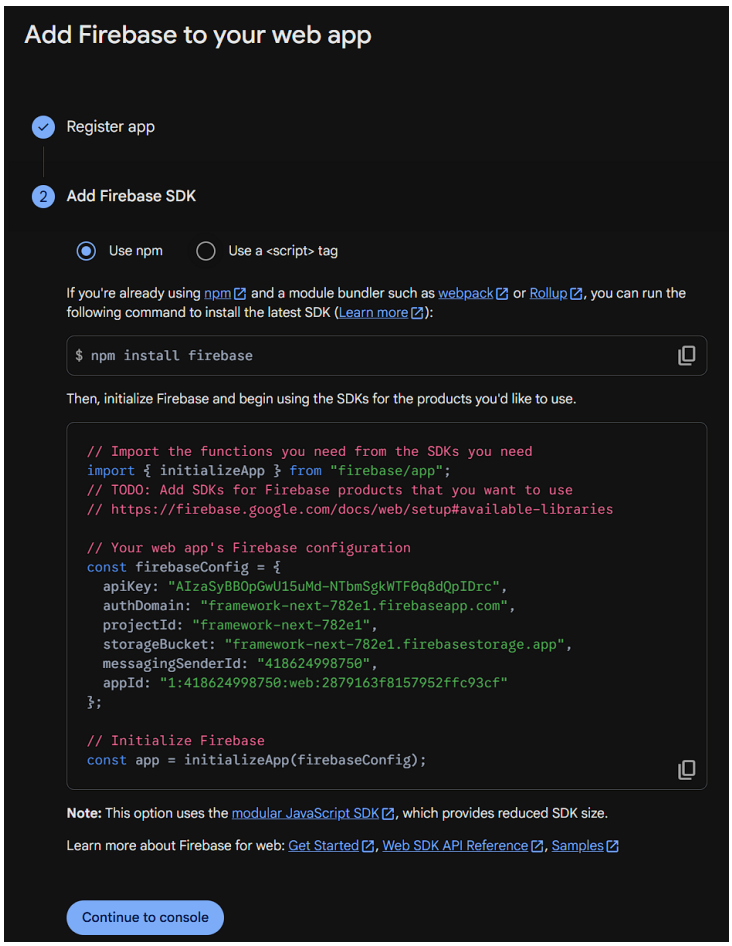
```tsx
import { useRouter } from "next/router";
import { useEffect, useState } from "react";

type ProductType = {
  id: string;
  nama: string;
  harga: number;
  ukuran: string;
  warna: string;
};

const kategori = () => {
  const [isLogin, setIsLogin] = useState(false);
  // const [push] = useRouter();
  const [products, setProducts] = useState<ProductType[]>([]);

  // useEffect(() => {
  //   if (!isLogin) {
  //     push("/auth/login");
  //   }
  // }, []);

  useEffect(() => {
    fetch("/api/produk")
      .then((response) => response.json())
      .then((responsedata) => {
        console.log("Data produk:", responsedata.data);
        setProducts(responsedata.data);
      })
      .catch((error) => {
        console.error("Error fetching produk:", error);
      });
  }, []);

  return (
    <div>
      <h1>Daftar Produk</h1>
      <div>
        {products.map((product: ProductType) => (
          <div key={product.id}>
            <h2>{product.nama}</h2>
            <p>Harga: {product.harga}</p>
            <p>Ukuran: {product.ukuran}</p>
            <p>Warna: {product.warna}</p>
          </div>
        ))}
      </div>
    </div>
  );
};

export default kategori;
```

2	Jalankan browser http://localhost:3000/produk

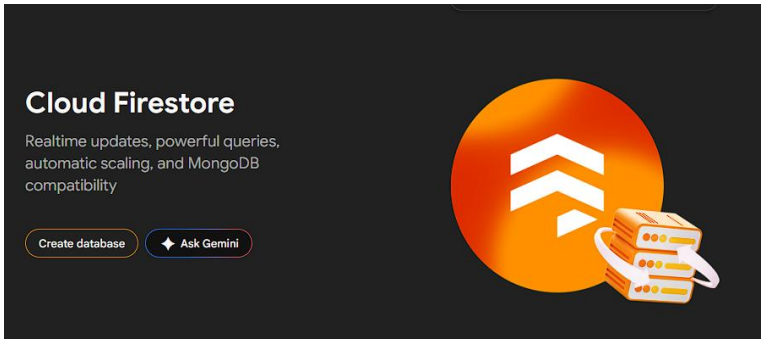
	 Tampilan Halaman Produk

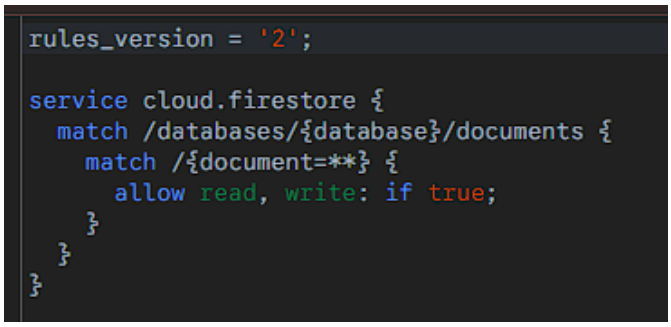
Langkah 4 – Setup Firebase

Langkah	Jawaban/Deskripsi
1	Buka Firebase Go To Console 
2	Buat project baru 

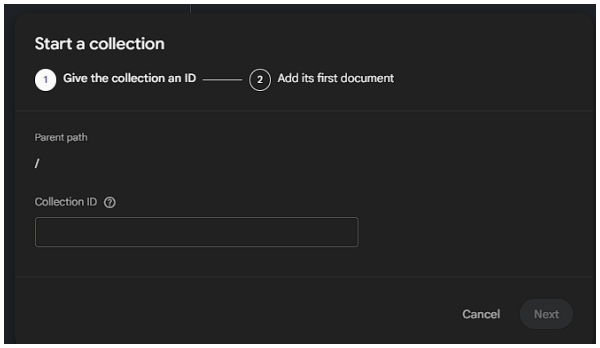
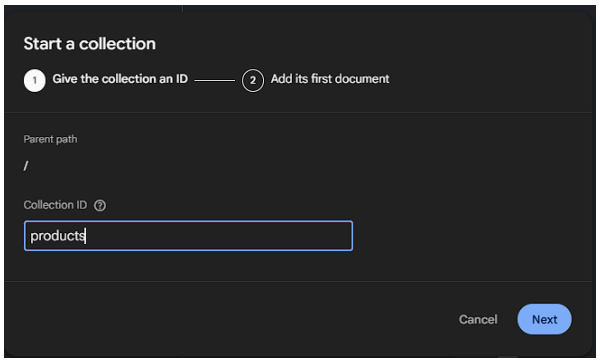
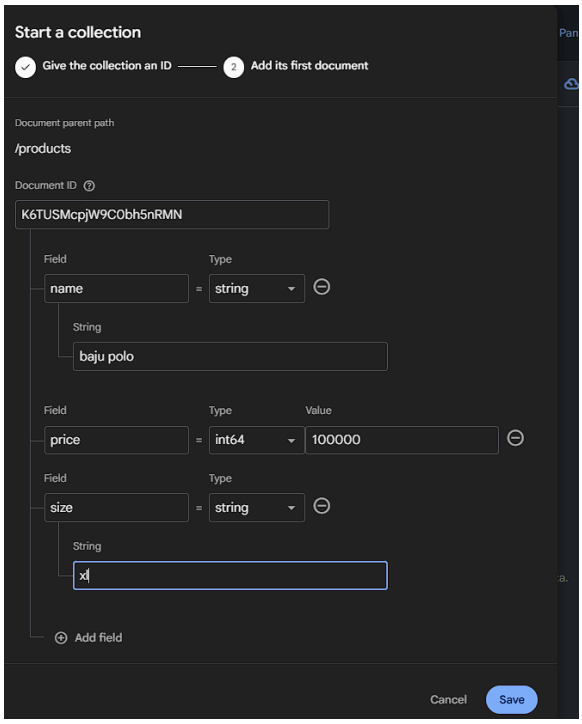


3 Aktifkan Firestore Database





```
rules_version = '2';

service cloud.firestore {
  match /databases/{database}/documents {
    match /{document=**} {
      allow read, write: if true;
    }
  }
}
```

4	Buat collection: 
5	Products 
6	Tambahkan field: 

	Berhasil Menambahkan Database dan juga berhasil menambahkan Produk ke dalam Database tersebut

Langkah 5 – Install Firebase

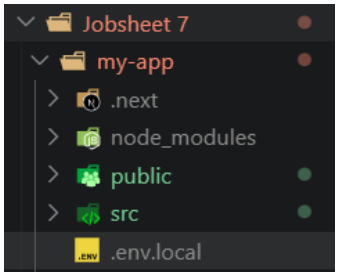
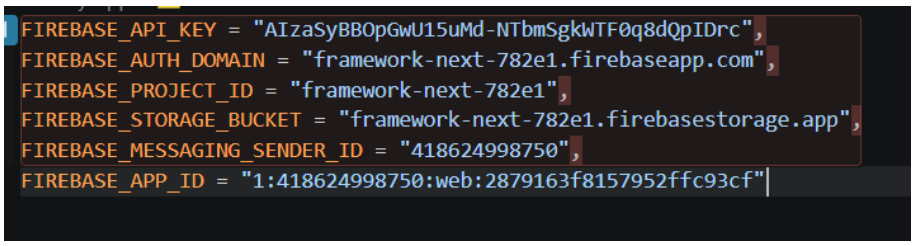
Langkah	Jawaban/Deskripsi

1	Npm install firebase
	`PS D:\Matkul\Semester 6\FrameWork\praktikum\Pemrograman-Berbasis-Framework\Jobsheet 7\my-app> npm install firebase`
	`added 143 packages, and audited 493 packages in 9m`
	`179 packages are looking for funding`
	`  run `npm fund` for details`
	`4 vulnerabilities (2 moderate, 2 high)`
	`To address issues that do not require attention, run:`
	`  npm audit fix`
	`To address all issues, run:`
	`  npm audit fix --force`
	`Run `npm audit` for details.`

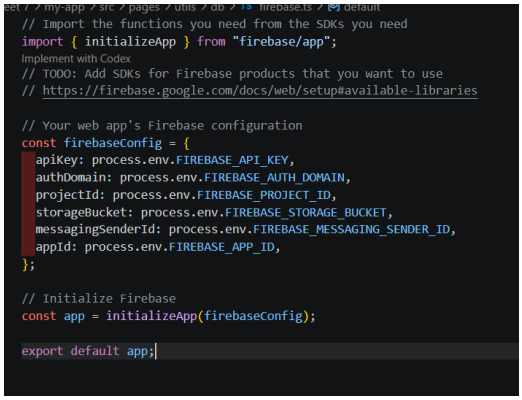
2	Buat folder dan file ts pada pages utlis/db/firebase.ts utils\db TS firebase.ts U

3	Copy paste yang ada pada kotak merah ke file firebase.ts
	`import { initializeApp } from "firebase/app";`
	`Implement with Codex`
	`// TODO: Add SDKs for Firebase products that you want to use`
	`// https://firebase.google.com/docs/web/setup#available-libraries`
	`// Your web app's Firebase configuration`
	`const firebaseConfig = {`
	`  apiKey: "AIzaSyBBOpGwU15uMd-NTbmSgkWTF0q8dQpIDrc",`
	`  authDomain: "framework-next-782e1.firebaseapp.com",`
	`  projectId: "framework-next-782e1",`
	`  storageBucket: "framework-next-782e1.firebasestorage.app",`
	`  messagingSenderId: "418624998750",`
	`  appId: "1:418624998750:web:2879163f8157952ffc93cf"`
	`};`
	`// Initialize Firebase`
	`const app = initializeApp(firebaseConfig);`
	Berhasil menginisialisasi firebase dengan projek

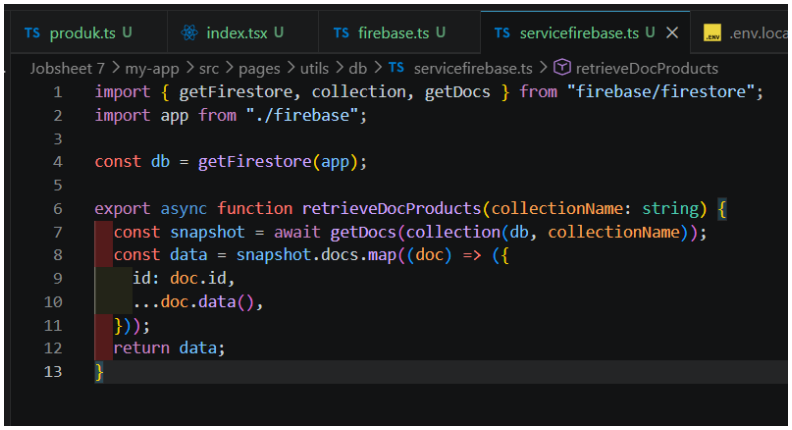
Langkah 6 – Konfigurasi Environment Variable agar credensial firebase tidak dapat dilihat saat dipush di repository

Langkah	Jawaban/Deskripsi
1	Buat file: .env.local 
2	Modifikasi file env  Berhasil melakukan modifikasi file .env

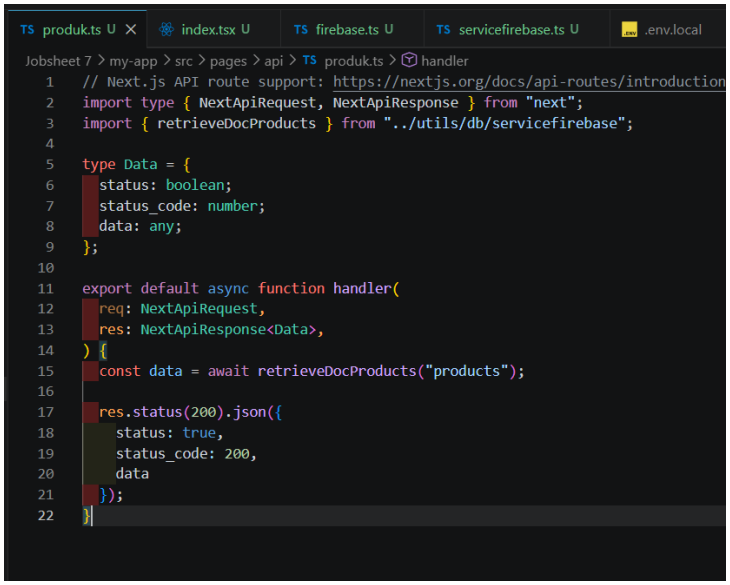
Langkah 7 - Konfigurasi Firebase

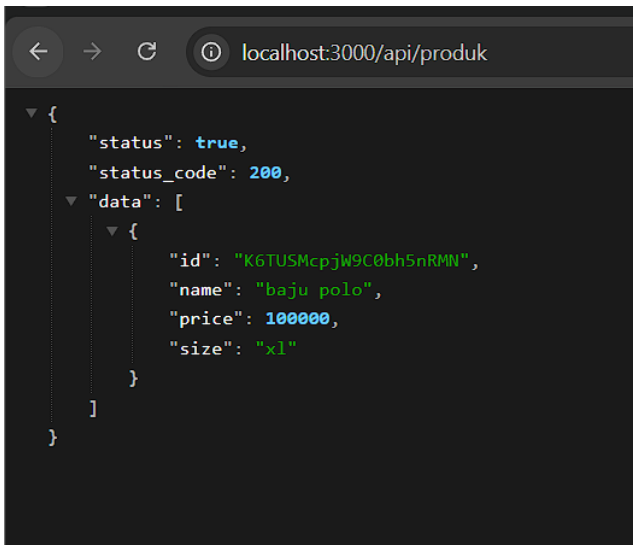
Langkah	Jawaban/Deskripsi
1	Modifikasi firebase.ts  Melakukan modifikasi firebase.ts dengan memanggil variable .env

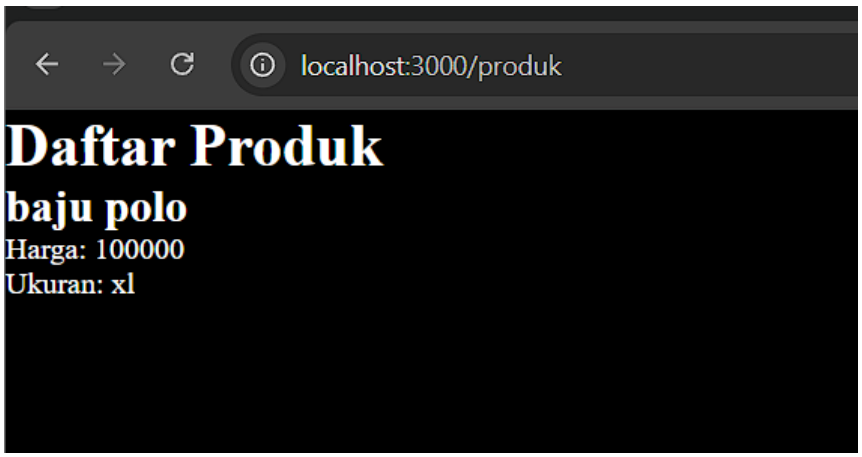
Langkah 8 - Ambil Data dari Firestore

Langkah	Jawaban/Deskripsi
1	Buat file  Menambahkan file servicedatabase dan mengisinya

Langkah 9 - API Mengambil Data Firebase

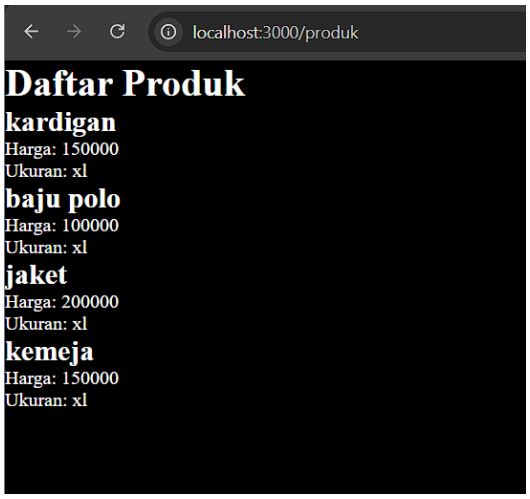
Langkah	Jawaban/Deskripsi
1	Edit pages/api/product.js: 

2	Jalankan browser http://localhost:3000/api/produk 
3	Modifikasi index.ts pada produk sesuaikan nama typenya dan db nya

	 Berhasil menampilkan produk dari database ke dalam website.

F. Tugas Praktikum

Tugas 1

Langkah	Jawaban/Deskripsi
1	Tambahkan minimal 3 data produk di Firestore
2	Pastikan data tampil di halaman produk 

Tugas 2

Langkah	Jawaban/Deskripsi
1	Tambahkan field baru: Category

Tugas 3

Langkah	Jawaban/Deskripsi
1	Tambahkan tombol Refresh Data

2	Gunakan fetch ulang tanpa reload halaman
	
	`onClick={fetchProducts}`
	`disabled={isLoading}`
	`style={{`
	`padding: "8px 16px",`
	`marginBottom: "16px",`
	`backgroundColor: isLoading ?`
	`"#ccc" : "#007bff",`
	`color: "white",`
	`border: "none",`
	`borderRadius: "4px",`
	`cursor: isLoading ? "not-allowed"`
	`: "pointer",`
	`}}`
	`>`
	`{isLoading ? "Loading..." :`
	`"Refresh Data"}`
	`</button>`

G. Pertanyaan Evaluasi

Tugas 3

Langkah	Jawaban/Deskripsi
1	Apa fungsi API Routes pada Next.js? Untuk membuat API backend langsung di dalam projek Next.js. Jadi setiap file di folder pages/api atau app/api otomatis menjadi endpoint
2	Mengapa .env.local tidak boleh di-push ke repository? Karena file tersebut berisi secret key (API Key, password, dll). Jika di-push, orang lain bisa melihat dan menyalahgunakannya.
3	Apa perbedaan data statis dan data dinamis? **Statis**: Data diambil saat build, cepat, dan tidak berubah. **Dinamis**: Data diambil saat user membuka halaman, selalu up-to-date.
4	Mengapa Next.js disebut framework fullstack? Karena Next.js bisa menangani frontend (React) dan backend (API, database, server logic) dalam satu project yang sama.